

BHOPAL, India

WMO: 426670

Lat: 23.287N

Lon: 77.337E

Elev: 1719

StdP: 13.81

Time Zone: 5.50 (IND)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) 48.5	(c) 51.0	(d) 28.5	(e) 24.0	(f) 75.7	(g) 31.9	(h) 28.1	(i) 74.2	(j) 14.3	(k) 73.4	(l) 12.4	(m) 70.9	(n) 3.7	(o) 50	(p) 0.372

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
5	22.7	107.6	71.5	105.5	70.7	103.3	70.4	79.4	88.2	78.6	86.9	77.9	85.8	10.6	290

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
77.4	152.1	82.3	76.9	149.2	81.6	76.2	145.9	80.9	44.7	88.5	43.8	86.4	43.0	85.9	86.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 19.4	(b) 17.4	(c) 14.9	(e) 43.9	(f) 110.6	(g) 2.4	(h) 1.9	(i) 42.2	(j) 112.0	(k) 40.8	(l) 113.1	(m) 39.4	(n) 114.2	(o) 37.7	(p) 115.6		
(4) 19.4	(5) 17.4	(6) 14.9	(7) 43.9	(8) 110.6	(9) 2.4	(10) 1.9	(11) 42.2	(12) 112.0	(13) 40.8	(14) 113.1	(15) 39.4	(16) 114.2	(17) 37.7	(18) 115.6		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	78.3	64.2	69.9	79.0	87.8	92.8	88.3	80.5	78.6	80.2	78.7	72.6	66.2
	DBStd	9.36	4.51	5.00	5.09	3.79	3.67	5.79	3.62	2.31	2.47	3.02	3.36	4.00
	HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD65	119	71	14	0	0	0	0	0	0	0	0	0	34
	CDD50	10313	441	558	899	1133	1328	1148	944	887	906	889	678	502
	CDD65	4956	47	151	434	683	863	698	479	422	456	424	228	71
	CDH74	59844	474	1524	5284	10328	14105	9855	4375	3027	4027	4235	1947	663
	CDH80	31886	71	531	2873	6607	9739	5957	1410	630	1434	1925	603	106
(14) Wind	WSAvg	6.8	4.8	5.4	6.1	7.7	9.8	10.1	9.6	8.8	6.6	4.4	3.7	4.1
(15) Precipitation	PrecAvg	44.6	0.5	0.4	0.4	0.2	0.5	6.3	16.7	11.2	6.8	1.1	0.4	0.3
	PrecMax	75.0	2.6	1.9	3.5	1.1	2.6	18.1	30.8	34.5	20.6	5.0	4.4	4.5
	PrecMin	31.3	0.1	0.0	0.0	0.0	0.0	0.9	3.8	1.3	1.2	0.0	0.0	0.0
	PrecStd	9.9	0.6	0.6	0.8	0.2	0.7	4.3	5.5	7.1	4.6	1.3	0.9	0.9
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	84.9	92.3	101.4	107.4	110.4	109.8	96.3	89.9	94.2	95.5	90.2	85.7
		MCWB	62.9	64.9	65.0	69.2	71.1	74.1	75.3	77.6	72.9	68.4	66.3	63.6
	2%	DB	82.0	89.0	98.5	105.3	108.4	106.3	92.5	87.4	91.8	93.1	87.6	82.8
		MCWB	61.9	63.4	64.9	68.2	71.2	73.6	75.9	77.3	74.1	69.5	65.8	62.6
	5%	DB	79.4	86.0	96.1	103.4	106.6	103.1	89.8	85.5	89.6	91.1	85.7	80.3
		MCWB	60.8	62.6	65.1	68.1	71.1	74.0	76.5	76.9	74.6	69.4	65.2	61.5
	10%	DB	76.3	83.1	93.4	101.1	104.5	99.6	87.2	83.7	87.4	89.3	83.7	77.8
		MCWB	59.3	61.8	64.6	67.7	70.8	74.3	76.5	76.6	75.2	68.8	64.8	60.5
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	66.8	68.6	72.1	77.3	80.5	80.1	80.7	79.9	79.6	77.7	73.8	68.0
		MCDB	78.9	83.7	90.6	101.2	100.3	93.0	87.5	85.7	85.7	86.7	82.7	79.2
	2%	WB	64.4	66.6	69.4	73.3	77.3	78.8	79.2	78.8	78.3	76.3	71.6	65.9
		MCDB	76.4	82.1	89.8	98.4	98.0	91.5	85.7	84.5	84.8	84.9	79.3	77.6
	5%	WB	62.7	65.0	67.7	71.1	75.5	77.9	78.4	78.0	77.4	74.9	69.3	63.8
		MCDB	74.7	80.3	88.6	96.5	96.5	90.3	84.4	83.0	84.0	82.9	78.8	75.2
	10%	WB	61.0	63.5	66.1	69.4	74.0	77.1	77.6	77.2	76.7	73.5	67.3	62.0
		MCDB	72.6	78.7	88.0	95.6	95.3	89.2	83.2	81.7	83.2	81.6	78.7	73.8
(35) Mean Daily Temperature Range	5% DB	MDBR	22.7	24.1	25.8	26.2	22.7	17.6	10.1	8.8	13.6	21.4	23.3	23.4
		MCDBR	26.3	27.4	28.7	28.1	24.0	21.4	15.3	12.4	18.1	23.8	24.7	26.2
		MCWBR	11.2	10.3	10.2	10.0	8.6	7.1	5.1	5.1	5.9	8.1	9.9	11.0
	5% WB	MCDBR	21.7	23.1	25.7	25.5	21.6	17.3	11.2	10.3	12.7	16.8	19.9	22.3
		MCWBR	10.6	10.5	10.9	11.2	8.9	6.3	5.3	5.1	5.7	6.9	9.7	10.8
(40) Clear-Sky Solar Irradiance	taub		0.435	0.439	0.474	0.529	0.579	0.649	0.620	0.564	0.518	0.505	0.503	0.458
	taud		2.063	2.089	2.009	1.912	1.843	1.751	1.826	1.982	2.077	2.009	1.941	2.014
	Ebn at Noon		258	267	264	251	237	220	226	239	248	244	234	244
	Edh at Noon		48	49	56	63	67	73	68	58	52	53	53	48
(44) All-Sky Solar Radiation	RadAvg		1372	1669	1999	2224	2246	1778	1208	1181	1552	1679	1441	1336
	RadStd		98	76	87	75	54	191	185	151	168	92	85	49

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	+4.36	N/A	+0.59	+0.79	N/A	N/A	N/A	N/A
(47) Regional (2 neighbors)	N/A	N/A	+4.36	N/A	+0.59	+0.79	N/A	N/A	N/A	N/A

Nomenclature: See separate page